(4) Test of Hypothesis about Coefficient of Correlation

Case 1: Testing the hypothesis when the population coefficient of correlation equals zero, *i.e.,* $H_0 : \rho = 0$

Here the null hypothesisis that there isno correlation in the population, *i.e.,* $H_0 : \rho = 0$. The population coefficient of correlation ρ measuresthe degree of relationship between the variables.When $\rho = 0$, there is no statistical relationship between the variables.In order to test this hypothesis, it is necessary to know the sample coefficient of correlation *r* (which is the best estimate of ρ). The appropriate test statistic to be used here is given by:

$$t = \frac{r}{\sqrt{1-r^2}} \times \sqrt{n-2}$$

which followst-distribution with *n* - 2 degrees of freedom.

If the computed value of *t* is -greater than the table value of *t,* the null hypothesis is rejected which indicates that sample data provides sufficient evidence to indicate that $\rho \neq 0$. Hence,it can be concluded that there is alinear relationship between the variables.

Example 7

In a study of the relationship between expenditure (X) and annual salesvolume *(Y,*a sample of 10 firms yieldedthe coefficient ofcorrelation r = 0.93. Can we conclude on the basis of this data that X and Y are linearly related?

Solution

Thenull hypothesis is $H_0 : \rho = 0$, i.e.,there is no relationship between two variables. Using the t-test;

$$t = \frac{r}{\sqrt{1-r^2}} \times \sqrt{n-2} = \frac{0.93}{\sqrt{1-(0.93)^2}} \times \sqrt{10-2} = \frac{0.93}{\sqrt{0.14}} \times \sqrt{8} = \frac{0.93 \times 2.828}{0.374} = 7.03$$

The degrees of freedom or v = n – 2 = 10 – 2 = 8.

The table value of *t at* 5%level of significance for 8 *d.f. is 2.306*.Since the computed value ismuch greater value, thenull hypothesis is rejected.Hence,it maybe concluded that X and Y are linearly related.

Case 2: Testing the hypothesis when the population coefficient of correlation of other value than zero, *i.e.,* $H_0 : \rho = \rho_0$

In this case when $\rho \neq 0$, the test based on t-distribution will not be appropriate. In testing the hypothesis, the use of Fisher's z-transformation will be applicable.Here,*r* is transformed into zby

$$z = \frac{1}{2}\log_e\left[\frac{1+r}{1-r}\right]$$

Here, $\log_e$ is a *natural* logarithm. Common logarithms may be shifted to natural logarithms by multiplying by the factor 2.3026, i.e.,

$\log_e X = 2.3026 \log_{10} X$, where X isa positive integer.

Since $\frac{1}{2}(2.3026) = 1.1513$, the transformation formula may be used as :

$$z = \frac{1}{2}\log_e\left[\frac{1+r}{1-r}\right] = 1.1513\log_{10}\left[\frac{1+r}{1-r}\right]$$

Now, it can be shown that Z is normally distributed with mean

$z_\rho = \frac{1}{2}\log_e\left[\frac{1+\rho}{1-\rho}\right]$ and standard deviation , $\sigma_z = \frac{1}{\sqrt{n-3}}$

Therefore, to test the null hypothesis that $\rho = \rho_0$,the test statistic would be:

$$z = \frac{z - z_\rho}{\sigma_z}$$

which follows approximately the standard normal distribution. This test is more appropriate if sample size is large. The approximation is reasonably good if the sample size is at least 10.

Case 3 : Testing the hypothesis for the difference between two independent correlation coefficients

To test the hypothesis of two correlation coefficients derived from two separate samples, we have to compare the difference of the two corresponding values of zwith the standard error of that difference. In this case, the formula used will be:

$$z = \frac{z_1 - z_2}{\sigma_{z_1 - z_2}} = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}}}$$ where:

$$z_1 = \frac{1}{2}\log_e\left[\frac{1+r_1}{1-r_1}\right] = 1.1513\log_{10}\left[\frac{1+r_1}{1-r_1}\right] \text{ and } z_2 = \frac{1}{2}\log_e\left[\frac{1+r_2}{1-r_2}\right] = 1.1513\log_{10}\left[\frac{1+r_2}{1-r_2}\right].$$

If the absolute value of this statistic is greater than 1.96, the difference will be significant at 5 % level.

Example 8

The following data give sample size:

Sample size	Value of r
5	0.87
12	0.56

Test the significance of the difference between two values using Fisher's z-transformation.

Solution

Let the null hypothesis be that the experiment provides no evidence that the samples aredrawn from the same population.Applying z-test:

$$z = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}}}$$ where:

$$z_1 = 1.1513 \log_{10}\left[\frac{1+r_1}{1-r_1}\right] = 1.1513 \log_{10}\left[\frac{1+0.87}{1-0.87}\right] = 1.1513 \log_{10}\left[\frac{1.87}{0.13}\right] = 1.1513 \times 1.579 = 1.82$$

$$z_2 = 1.1513 \log_{10}\left[\frac{1+r_2}{1-r_2}\right] = 1.1513 \log_{10}\left[\frac{1+0.56}{1-0.56}\right] = 1.1513 \log_{10}\left[\frac{1.56}{0.44}\right] = 1.1513 \times 0.5495 = 0.63$$

$$z = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}}} = \frac{1.82 - 0.63}{\sqrt{\frac{1}{5-3} + \frac{1}{12-3}}} = \frac{1.19}{\sqrt{0.5 + 0.1}} = \frac{1.19}{\sqrt{0.6}} = \frac{1.19}{0.77} = 1.54$$

Since the computed value of zis lessthan the table value of z at 5% level of significance = 1.96, therefore,the null hypothesis is accepted.Hence,the experiment provides no evidence against the hypothesis that the samples are drawn from the same population.

Example 9

Suppose we want to test whether $r = 0.884$ for a pair of 20 observations is significantly different from a hypothesized value $\rho = 0.92$.

Solution

We transform *r* into zby

$$z = 1.1513 \log_{10}\left[\frac{1+0.884}{1-0.884}\right] = 1.3938$$

The distribution of zis approximately normal around the hypothesized value

$\rho_0 = 0.92 = z_{\rho_0}$, where

$$z_{\rho_0} = 1.1513 \log_{10}\left[\frac{1+0.92}{1-0.92}\right] = 1.5890$$

The distribution of z hasa standard deviation

$$\sigma_z = \sqrt{\frac{1}{n-3}} = \sqrt{\frac{1}{20-3}} = \sqrt{\frac{1}{17}} = 0.2425$$

Therefore, the statistic is

$$z = \frac{1.3938 - 1.5890}{0.2425} = -0.80$$

From the table of areas for the normal curve, we find that at = 5% level of significance, z = 1.96. Since computed $|z| = 0.8 < 1.96$, we accept H_0 and conclude that $\rho = 0.92$.

The F - Distribution

The *F-distribution* is named in honour of R.A. Fisher who *first studied it* in 1922 is usually defined in terms of the ratio of the variances of two normally distributed quantity

$$\frac{s_1^2 / \sigma_1^2}{s_2^2 / \sigma_2^2}$$

is distributed as *F-distribution* with $v_1 = n_1 - 1$ *and* $v_2 = n_2 - 1$ degrees of freedom, where

$s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1}$ is the unbiased estimator of σ_1^2 and $s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1}$ is the unbiased estimator of σ_2^2.

If $\sigma_1^2 = \sigma_2^2$,then the statistic;

$$F = \frac{s_1^2}{s_2^2}$$

follows *F-distribution* with $v_1 = n_1 - 1$ *and* $v_2 = n_2 - 1$ degrees of freedom.

The F-distribution sometimes is also called *VarianceRatio*distribution.

The F-distribution depends on the degrees of freedom v_1 for the numerator and v_2 for the denominator.Therefore, the parameters for F-

distribution are v_1 *and* v_2.For different values of v_1 *and* v_2, we shall have different distributions.

The probability density function of *F-distribution* is given by:

$$f(F) = Y_0 \frac{F^{\frac{v_1}{2}-1}}{\left[1+\frac{v_1}{v_2}\right]^{(v_1+v_2)/2}}, \quad 0 \le F \le \infty$$ where Y_0 is a constant depending on values v_1 *and* v_2 such that the area under the curve is unity. A typical F-distribution is given below:

Some of the important properties of the F-distribution are given below:

1) The F-distribution is positively skewed and its skewness decreases with increase in v_1 *and* v_2.

(2) The value of *F* must always be positive or zero since variances are squares and can never assume negative values. Its value will always lie between 0 and ∞.

(3) The shape of the F-distribution depends upon the number of degrees of freedom.

(4) The areas in the left-hand side of the distribution can be found by taking the reciprocal of F values corresponding to the right-hand side,when the number of degrees of freedom in the numerator and in the denominator are interchanged. This is also known as *reciprocal property and can be expressed as:*

$$F_{1-\alpha}(v_1, v_2) = \frac{1}{F_{\alpha}(v_2, v_1)}$$

where the symbols have their usual meanings. This property is of great help when we want to know the lower tail *F* values from corresponding upper tail *F* values which are given in the appendix.

Testing of Hypothesis for Equality of two Variances

The test of equality of two population variances is based on the variances in two independently selected random samples drawn from two normal populations. Under the null hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$;

$F = \frac{s_1^2 / \sigma_1^2}{s_2^2 / \sigma_2^2}$ reduces to $F = \frac{s_1^2}{s_2^2}$.

which follows F-distribution with v_1 *and* v_2 degrees of freedom. It is convenient to place larger sample variance in the numerator for computational purpose. If we do so, the ratio of the sample variance will be equal to or greater than one.

If the computed value of *F* exceeds the table value of F,we reject the null hypothesis, *i.e.,* the alternative hypothesis is accepted.

Example 10

Two sources of rawmaterials are under consideration bya company. Both sources seem to have similar characteristics but the company isnot sure about their respective uniformity. A sample of 10 lots from source *A* yields a variance of 225 and asample of 11 lots from source Byields avariance of 200.Is it likely that the variance of source Ais significantly greater than the variance of source Bat $\alpha = 0.01$?

Solution

Null hypothesisis $H_0 : \sigma_1^2 = \sigma_2^2$, i.e., the variances of source A and that of source Bare the same. The *F* statistic to be used here is:

$$F = \frac{s_1^2}{s_2^2} = \frac{225}{200} = 1.1$$

The table value of F for $v_1 = 9$ *and* $v_2 = 10$ at 1% level of significance is 4.94.

Since the computed value of *F* issmaller than the table value of F,the null hypothesis is accepted. Hence,the population variances of the two populations are same.

Confidence Interval for the Ratio of Two Variances

A $100(1-\alpha)$ per cent confidence interval for the ratio of the variances of two normally distributed populations is given by:

$$\frac{s_1^2 / s_2^2}{F_{1-\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2 / s_2^2}{F_{\alpha/2}(v_1, v_2)}$$

where the symbols have their usual meaning.